

Apex & Flow Explorer

IMPLEMENTATION ADDENDUM - top-level metadata explanation workspace.

Status: Working. The sidebar contains a dedicated **Apex & Flow Explorer** section with route-backed **Apex Classes**, **Apex Triggers**, and **Flows** subtabs. It reuses and expands the former Code & Automation Explainer.

Component and route inventory

Component	Responsibility	Implementation
Navigation	Top-level apexFlow section, three searchable tools, route history, deep link by explorerName.	src/sidepanel/state/routes.ts; src/sidepanel/App.tsx
Explorer UI	Responsive master-detail browser, search, Flow version picker, refresh, copy, source/metadata panel, handler links.	src/sidepanel/modules/development/CodeExplainer.tsx
Local analyzers	Apex/trigger structural parsing, line references, dependency detection, Flow element grouping and connector path.	src/shared/development/codeExplainer.ts
API and cache	Tooling queries and sObject reads through SalesforceClient; org-scoped definition cache in IndexedDB.	src/shared/api/client.ts; services/platform.ts
Shared UI	ConnectionGate, theme tokens, loading/error/empty states, buttons, pills, focus and responsive behavior.	src/sidepanel/components; src/sidepanel/styles.css

Common data flow and privacy

- **1. Select org:** Workspace context exposes the explicitly selected connected org. Changing org remounts the tool and changes the storage namespace.
- **2. List metadata:** Tooling Query returns only classes, triggers, and Flow versions visible to the connected Salesforce user.
- **3. Retrieve definition:** Tooling sObject GET retrieves Body for Apex or Metadata for the selected Flow version.
- **4. Explain locally:** Deterministic parsers inspect structural markers. No source or metadata is sent to an AI service or any non-Salesforce host.
- **5. Cache:** Results use code-explainer:{type}:{id}:{lastModified-or-version} inside the current org IndexedDB namespace. Refresh bypasses the cache.
- **6. Verify:** Users can inspect and copy the original source or metadata and compare it with cited methods, lines, or Flow element names.

Salesforce API surface

Purpose	Tooling API operation
List classes	SELECT Id, Name, Status, ApiVersion, LastModifiedDate FROM ApexClass ORDER BY Name
List triggers	SELECT Id, Name, Status, TableEnumOrId, ApiVersion, LastModifiedDate FROM ApexTrigger ORDER BY Name
List Flow versions	SELECT Id, MasterLabel, FullName, Status, VersionNumber, ProcessType, LastModifiedDate FROM Flow ORDER BY MasterLabel, VersionNumber DESC
Retrieve definition	GET /tooling/sobjects/ApexClass/{id}, ApexTrigger/{id}, or Flow/{id} under the negotiated REST API version.